Oxford College of Emory University

# Chapter 13: Fiscal Policy

## Government Purchases, Taxes, and Transfers

ECON_OX 112 · Principles of Macroeconomics

Guilherme Paiva Pinto

# The Federal Budget: FY2025

**Federal government only · October 2024–September 2025 · Billions of dollars**

| Revenues | $ billions |
|---|---|
| Individual income taxes | 2,656 |
| Payroll taxes | 1,748 |
| Corporate income taxes | 452 |
| Customs duties (tariffs) | 195 |
| Excise taxes | 106 |
| Estate and gift taxes | 29 |
| Federal Reserve remittances | 5 |
| Miscellaneous fees and fines | 42 |
| **Total revenues** | **5,235** |

| Outlays | $ billions |
|---|---|
| Social Security | 1,575 |
| Medicare (net) | 988 |
| Medicaid | 668 |
| Income security programs | 397 |
| Other mandatory (net) | 539 |
| Defense discretionary | 893 |
| Nondefense discretionary | 980 |
| Net interest | 970 |
| **Total outlays** | **7,010** |

**Budget balance: $5,235 − $7,010 = −$1,775 billion** (deficit)

A financing shortfall—not a measure of the effect on GDP.

Source: CBO, FY2025 Federal Budget; Revenues; 2026 Outlook. Components may not sum because of rounding. Medicare and other mandatory outlays are net of offsets.

# What Does Federal Spending Pay For?

> **Public services**
>
> National defense, federal courts, and law enforcement.

Source: CBO, Federal Budget and spending breakdowns, FY2025

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### Health care and local services

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Retirement and disability benefits, food assistance, and refundable tax credits.

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### Income support

Retirement and disability benefits, food assistance, and refundable tax credits.

### Interest on federal debt

Payments to lenders; these do not buy current goods or services.

Source: CBO, Federal Budget and spending breakdowns, FY2025

# Roadmap

## Policy goals and instruments

Purchases, household taxes, and transfers

## Demand and policy limits

Expansion, contraction, criticisms, and delays

## The multiplier

Initial spending and successive consumption rounds

## Stabilizers and budget balances

Automatic responses, deficits, and sustainability

# Fiscal Policy and Stabilization

**Definition**

Fiscal policy uses government purchases, taxes, and transfers to influence aggregate demand.

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### Stabilization goal

Support output and employment in a downturn; restrain excess demand and inflation pressure.

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Expansionary policy shifts AD right; contractionary policy shifts AD left.

# Fiscal Instruments

| Instrument | Example | Immediate channel |
|---|---|---|
| **Purchases (G)** | Road maintenance | A purchase of current output |
| **Transfers (TR)** | Unemployment benefits | Household disposable income |
| **Household taxes (T)** | Personal income taxes | Household disposable income |

Transfers are not G. Public infrastructure investment is G; I is private investment.

GDP accounting: BEA

# Fiscal Policy and Aggregate Expenditure

$$Y = \mathbf{C} + \mathbf{I} + \mathbf{G} + \mathbf{NX}$$

**C: Consumption**

Household taxes and transfers change disposable income.

Tax cuts or higher transfers can raise C.

**I: Private investment**

**G: Government purchases**

**NX: Net exports**

# Fiscal Policy and Aggregate Expenditure

$$Y = C + I + G + NX$$

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|---|---|
| **C: Consumption** | Household taxes and transfers change disposable income.<br>Tax cuts or higher transfers can raise C. |
| **I: Private investment** | Business tax incentives and expected sales affect investment.<br>Borrowing can also raise financing costs and weaken I. |
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# Fiscal Policy and Aggregate Expenditure

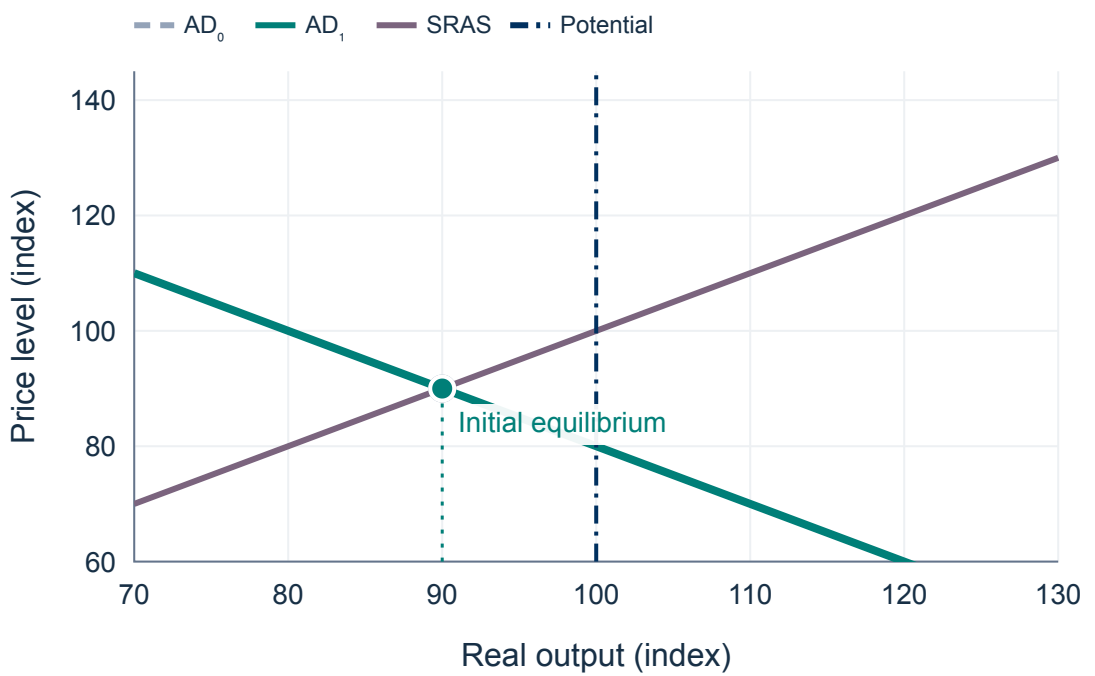
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| **NX: Net exports** | Higher domestic spending can increase imports.<br>Imports reduce NX, other things equal. |

# Expansionary Fiscal Policy

## Starting output = 90 · Potential output = 100

### Expansionary · AD increases



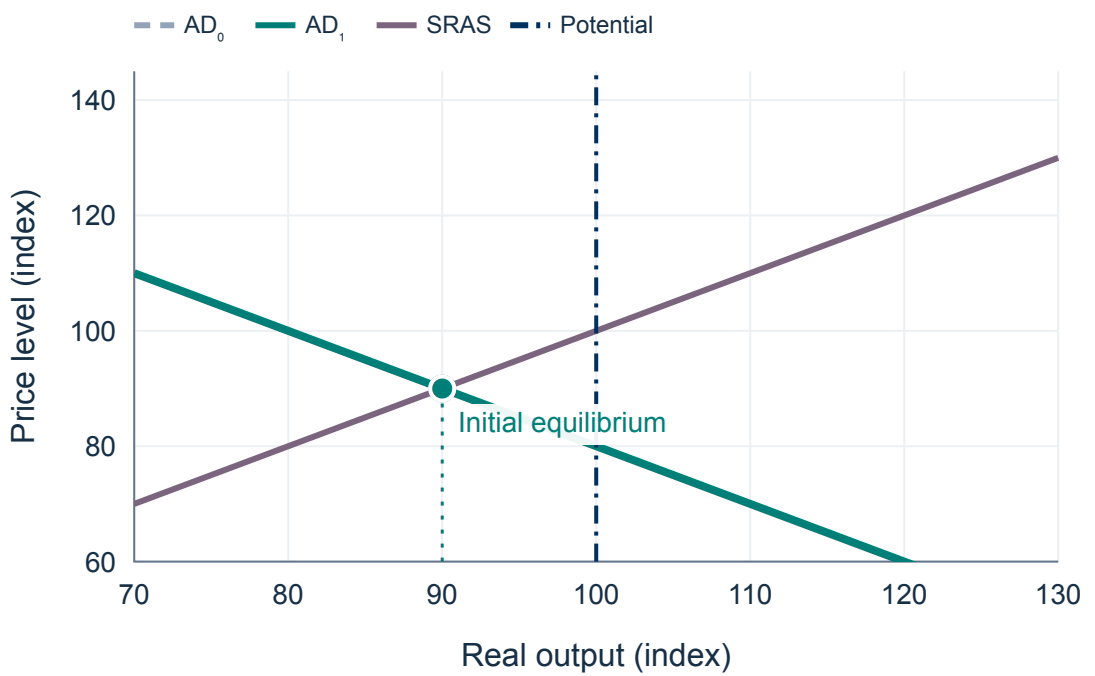

**Recessionary gap**

Output is below potential.

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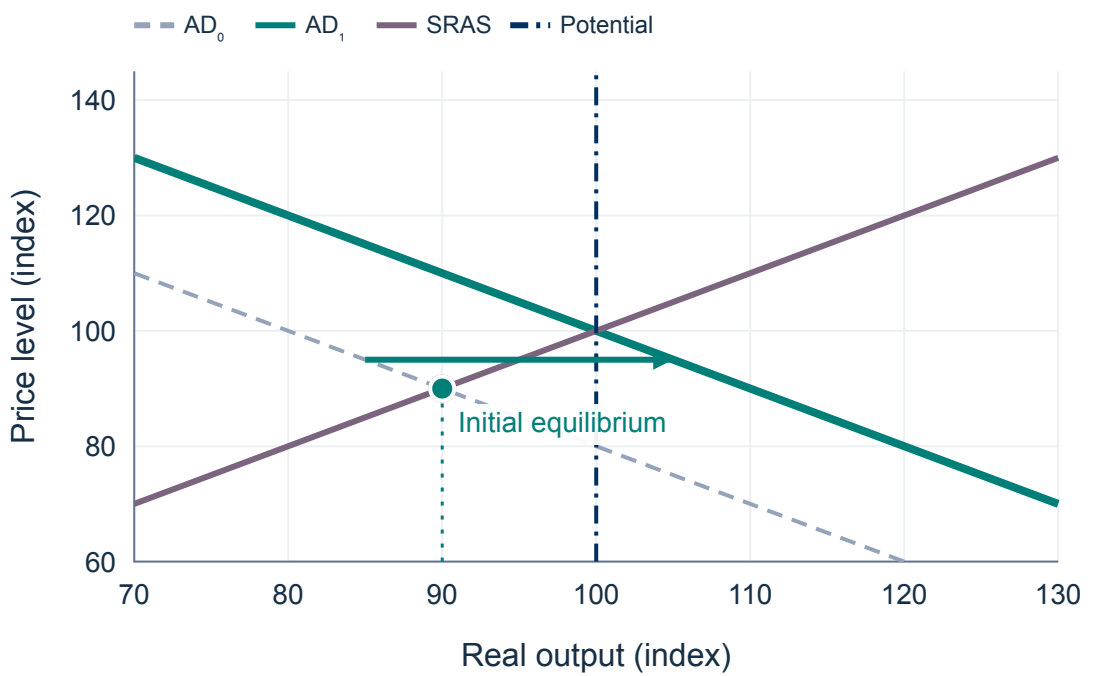
**Policy change**

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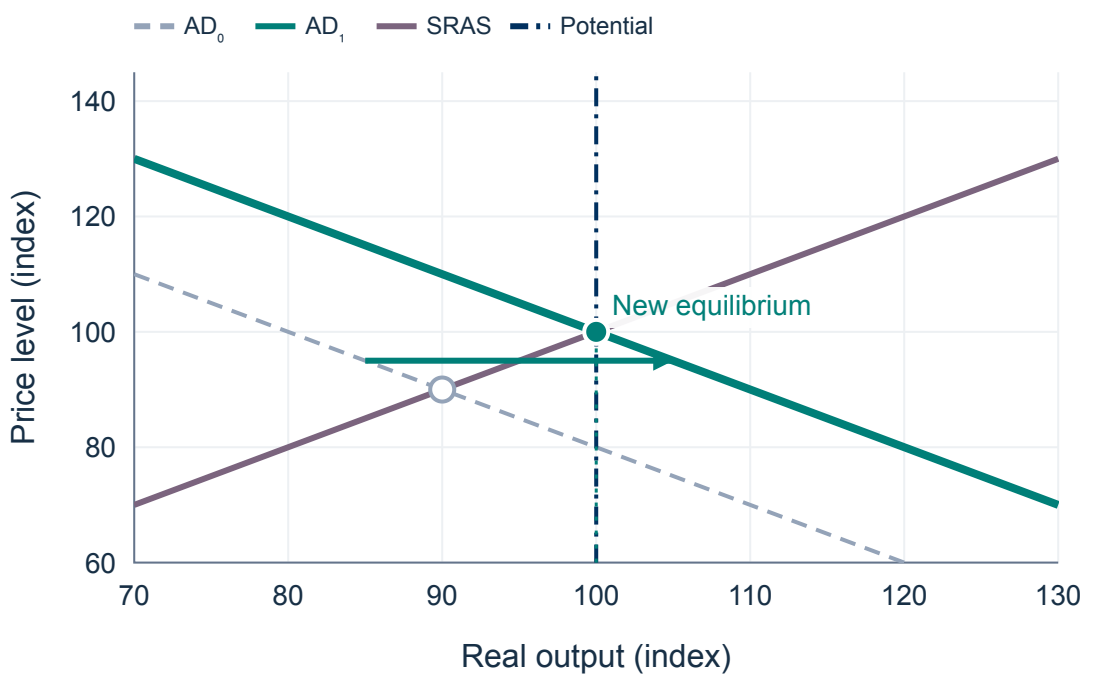
**Demand response**

AD shifts right; production rises.

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**Demand response**

AD shifts right; production rises.

**New equilibrium**

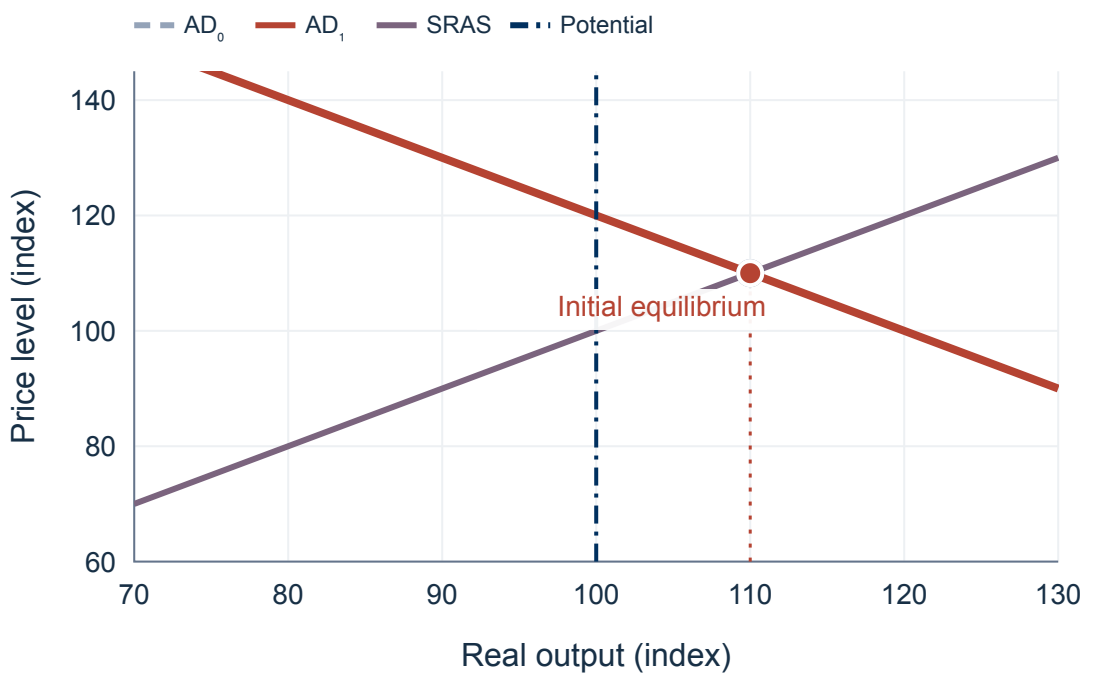
Output reaches 100; the price level rises.

## Expansionary alternatives: G ↑, household T ↓, or TR ↑.

# Contractionary Fiscal Policy

**Starting output = 110 · Potential output = 100**

## Contractionary · AD decreases



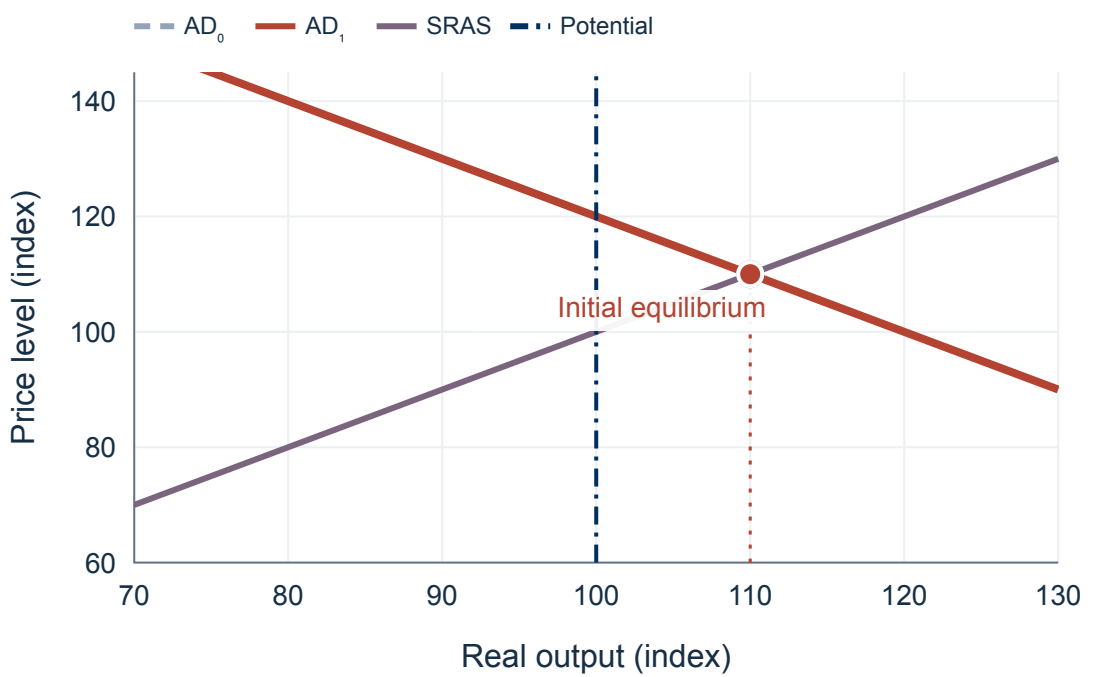

### Inflationary gap

Output is above potential.

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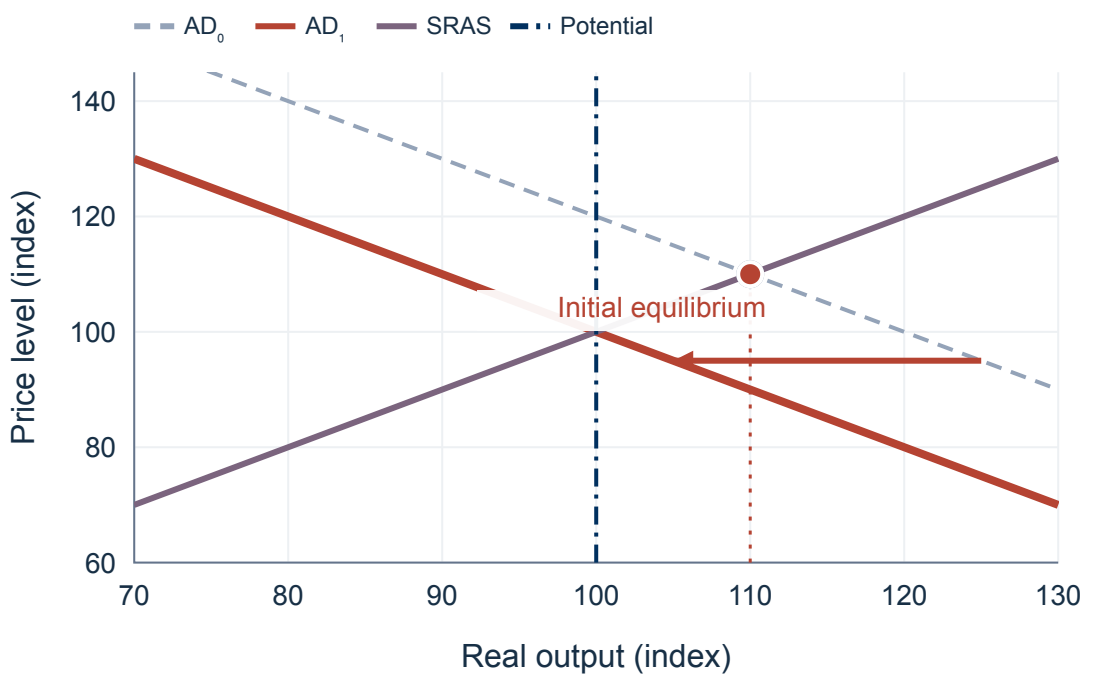
### Policy change

Government purchases decrease.

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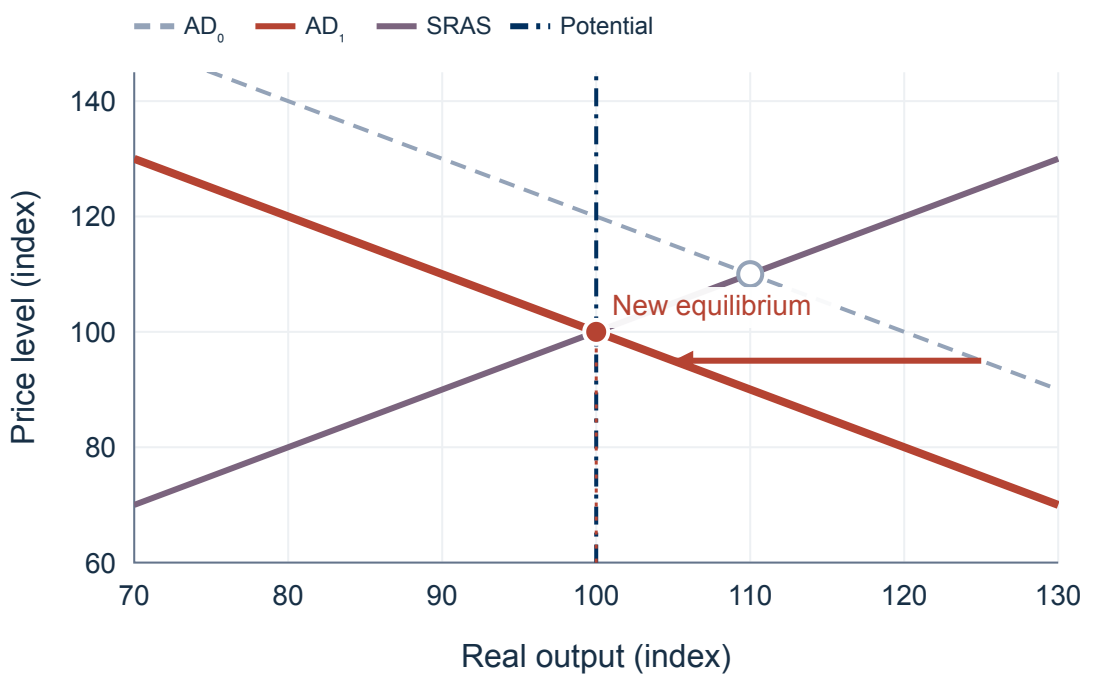
### Demand response

AD shifts left; production falls.

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### Policy change

Government purchases decrease.

### Demand response

AD shifts left; production falls.

### New equilibrium

Output returns to 100; the price level falls.

**Contractionary alternatives: G ↓, household T ↑, or TR ↓.**

# Fiscal Policy Checkpoint

Identify the policy direction and the immediate spending channel.

**A. Output is below potential; government repairs more roads.**

**B. Output is above potential; household income taxes rise.**

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Expansionary. Purchases directly increase G.

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Contractionary. Disposable income and consumption fall.

# Three Criticisms of Fiscal Expansion

| Criticism | Mechanism | Qualification |
| --- | --- | --- |
| **Investment crowding out** | Borrowing can raise interest rates and reduce I. | The size depends on economic and monetary conditions. |
| **Consumption offset** | | |
| **Ricardian equivalence** | | |

An offset can weaken a policy without eliminating its effect.

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| **Consumption offset** | Current tax financing can reduce disposable income and C. | The reduction need not cancel the extra G. |
| **Ricardian equivalence** | A tax cut may be saved for expected future taxes. | A conditional tax-timing argument, holding spending fixed. |

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# Fiscal Policy Lags: The 2008–2009 Response

## Recognition

Diagnose the shock.

**15 Sep 2008**

Lehman bankruptcy

**26 Jan 2009**

Recovery bill introduced

**17 Feb 2009**

ARRA signed

**May 2009**

$250 benefit payments

Authorization and delivery occur at different times.

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**Implementation**

Deliver purchases or payments.

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### Compare the total multiplier effect

Net output change: $\Delta Y$ = initial expenditure change × m.

---

The same m can give different output effects when initial spending differs.

# Government Purchases: Successive Spending Rounds

**ΔG = +$100 billion · MPC = 0.8 · Fixed price level**

## Government purchases

+$100 billion

Initial expenditure: ΔAE = +$100 billion.

### First spending

$100 billion

Fixed taxes and transfers; no imports or investment offsets.

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$64 billion + $51.2 billion + …

Net output effect: ΔY = $100bn × 5 = +$500bn

Spending-round factor: m = 1/(1 − 0.8) = 5

Fixed taxes and transfers; no imports or investment offsets.

# A Household Tax Cut

**Household taxes fall by $100 billion · MPC = 0.8**

> **Disposable income rises**
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> $\Delta T = -\$100$ billion; $\Delta Y_D = +\$100$ billion.

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### Keep the denominator clear

m = 400/80 = 5; output per $1 of tax cut = 400/100 = 4.

# Purchases, Tax Cuts, and Transfers

**MPC = 0.8 · Change one instrument at a time**

| Expansionary change | First channel | Initial expenditure |
|---|---|---|
| **G increases by $100 billion** | Government purchases | +$100 billion in G |
| **Household T falls by $100 billion** | | |
| **TR increases by $100 billion** | | |

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| **TR increases by $100 billion** | Disposable income +$100 billion | +$80 billion in C |

Same spending-round factor m = 5; the policy starts with $100 in G or $80 in C.

# Comparing Multiplier Effects

**MPC = 0.8 · m = 1/(1 − MPC) = 5 · Fixed lump-sum taxes**

| Policy change ($bn) | Initial ΔAE ($bn) | Common factor m | Net ΔY ($bn) | Net $\Delta S_g$ ($bn) |
|---|---|---|---|---|
| **ΔG = +100** | +100 in G | 5 | +500 | −100 |
| **ΔT = −100** | | | | |
| **ΔTR = +100** | | | | |

One policy change at a time; fixed prices, no imports or interest/investment offsets.

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| **ΔT = −100** | +80 in C | 5 | +400 | −100 |
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| **ΔTR = +100** | +80 in C | 5 | +400 | −100 |

## Output multiplier effect: ΔY = initial ΔAE × m

Same m = 5. Output per $1 of policy: purchases 5; tax cuts and transfers 4.

Net budget change: $\Delta S_g = \Delta T - \Delta G - \Delta TR$.

A $500bn output gap: ΔG = $500bn ÷ 5 = $100bn.

One policy change at a time; fixed prices, no imports or interest/investment offsets.

# Recipients and Policy Design

**An additional $100 of disposable income**

| Recipient | MPC | Initial consumption |
|---|---|---|
| **Household A** | 0.8 | $80 |
| **Household B** | 0.4 | |

# Recipients and Policy Design

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Consider spending, delivery speed, distribution, and the value of public purchases.

# Income Taxes and the Spending Multiplier

**MPC = 0.8 · Income-tax rate t = 25% · Fixed price level**

### Additional output

→ →

$\Delta Y = \$100$

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$\Delta Y = \$100$

$\rightarrow$

**Taxes and disposable income**

$\Delta T = \$25 \quad \Delta Y_D = \$75$

$\rightarrow$

# Income Taxes and the Spending Multiplier

**MPC = 0.8 · Income-tax rate t = 25% · Fixed price level**

| Additional output | | Taxes and disposable income | | Additional consumption |
|---|---|---|---|---|
| $\Delta Y = \$100$ | $\rightarrow$ | $\Delta T = \$25$ $\quad \Delta Y_D = \$75$ | $\rightarrow$ | $\Delta C = 0.8 \times \$75 = \$60$ |

Only \$60 reaches the next spending round, compared with \$80 under fixed taxes.

MPC remains 0.8; MPS remains 0.2.

# Income Taxes and the Spending Multiplier

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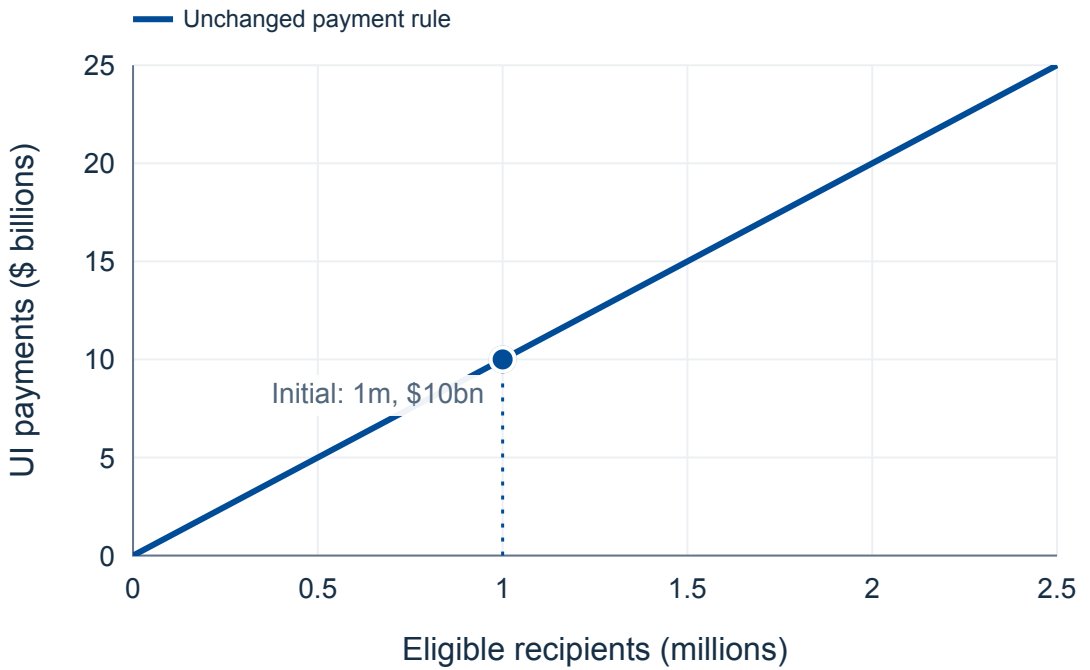
MPC remains 0.8; MPS remains 0.2.

$m = 1 / [1 - MPC(1 - t)]$

$m = 1 / [1 - 0.8(0.75)] =$ **2.5**, compared with **5** under fixed taxes.

# Automatic Stabilizers

## Unemployment insurance (UI): $10,000 per eligible recipient in this period



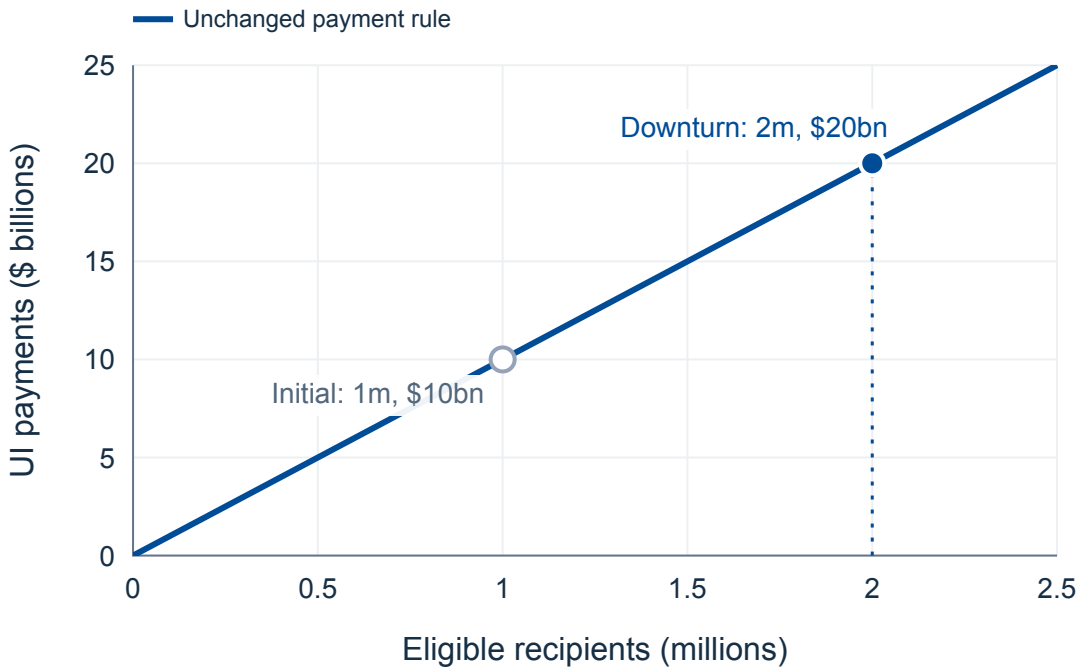

### Unemployment rises

More workers qualify under the same benefit rules.

Mechanism: CBO, Automatic Stabilizers (2024); graph values are model inputs.

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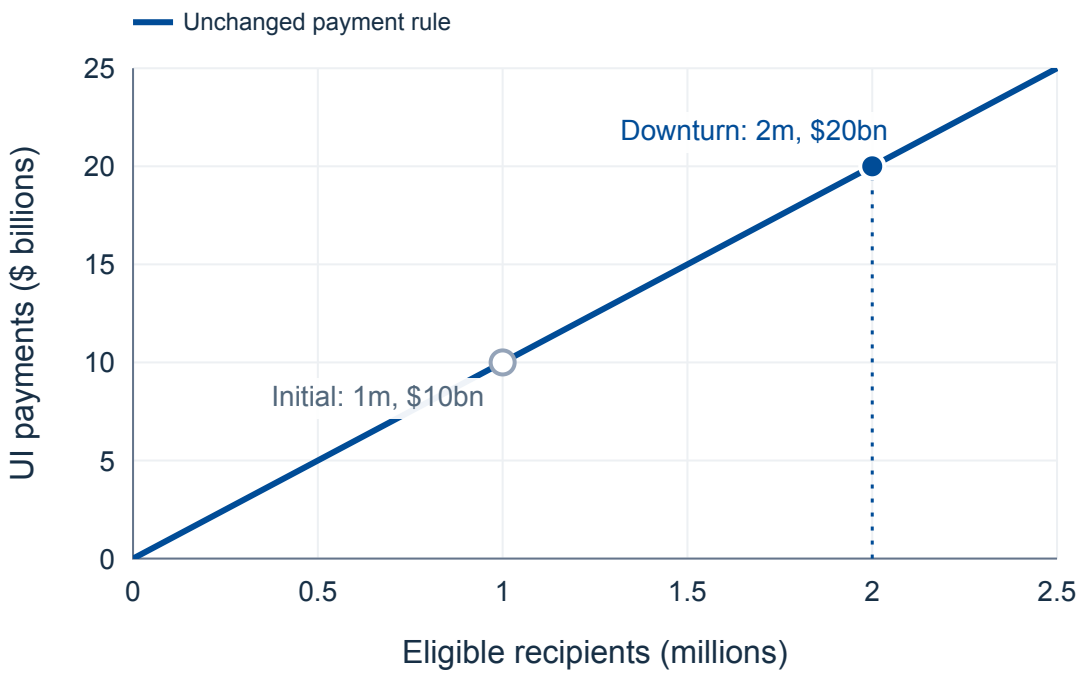
### Transfers rise automatically

1m → 2m recipients; UI payments rise $10bn → $20bn.

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# Automatic Stabilizers

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### Unemployment rises

More workers qualify under the same benefit rules.

### Transfers rise automatically

1m → 2m recipients; UI payments rise $10bn → $20bn.

### Consumption is cushioned

Benefits support disposable income and C; no new law is needed.

Income taxes also stabilize: falling income reduces tax payments under existing rules.

Mechanism: CBO, Automatic Stabilizers (2024); graph values are model inputs.

# Government Budget Balance

$$S_g = T - G - TR$$

T: tax revenue  G: purchases  TR: transfers

| Positive balance | Zero balance | Negative balance |
|---|---|---|
| Surplus | Balanced budget | Deficit |

Simplified budget: debt interest and other budget items omitted.

# Government Budget Balance

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T: tax revenue  G: purchases  TR: transfers

| Positive balance | Zero balance | Negative balance |
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| Surplus | Balanced budget | Deficit |

$T = \$500\text{bn} \cdot G = \$400\text{bn} \cdot TR = \$150\text{bn}$

$S_g = \$500\text{bn} - \$400\text{bn} - \$150\text{bn} = \mathbf{-\$50\text{bn}}$

Simplified budget: debt interest and other budget items omitted.

# Government Budget Balance

$$S_g = T - G - TR$$

T: tax revenue  G: purchases  TR: transfers

| Positive balance | Zero balance | Negative balance |
|---|---|---|
| Surplus | Balanced budget | Deficit |

$T = \$500\text{bn} \cdot G = \$400\text{bn} \cdot TR = \$150\text{bn}$

$S_g = \$500\text{bn} - \$400\text{bn} - \$150\text{bn} = \mathbf{-\$50\text{bn}}$

The balance alone does not reveal the effect on output.

Simplified budget: debt interest and other budget items omitted.

# Balanced-Budget Expansion

**$\Delta G = +\$100bn \cdot \Delta T = +\$100bn \cdot \Delta TR = \$0 \cdot MPC = 0.8 \cdot m = 5$**

| Component | Initial ΔAE ($bn) | Explanation |
|---|---|---|
| **Purchases rise 100** | +100 | A purchase of current output enters expenditure directly. |
| **Taxes rise 100** | | |
| **Net initial spending** | | |

Fixed lump-sum taxes and prices; unchanged transfers; no imports or interest/investment offsets.

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Purchases add $500bn; tax financing subtracts $400bn: net +$100bn.

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Net change in balance: $\Delta S_g = 100 - 100 - 0 =$ **$0**

The budget balance stays unchanged while output rises.

Fixed lump-sum taxes and prices; unchanged transfers; no imports or interest/investment offsets.

# The Budget and the Business Cycle

**Existing tax schedules and benefit rules · No new legislation**

| Budget item | Activity weakens | Activity strengthens |
|---|---|---|
| **Tax revenue** | Usually falls | Usually rises |
| **Unemployment benefits** | | |
| **Budget balance** | | |

Concept reference: CBO, Effects of Automatic Stabilizers on the Federal Budget (2024)

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| **Budget balance** | Moves toward deficit | Moves toward surplus |

Moving toward surplus can mean a smaller deficit.

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# Cyclically Adjusted Budget Balance

Estimated balance under current policies with real GDP at potential output.

| Measure | Balance ($ billions) |
|---|---|
| **Actual balance in recession** | −150 |
| **Estimated balance at potential** | |
| **Estimated cyclical contribution** | |

Potential output and cyclical responses are estimated.

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| **Estimated balance at potential** | −50 |
| **Estimated cyclical contribution** | −100 |

Actual = adjusted + cyclical

−150 = −50 + (−100)

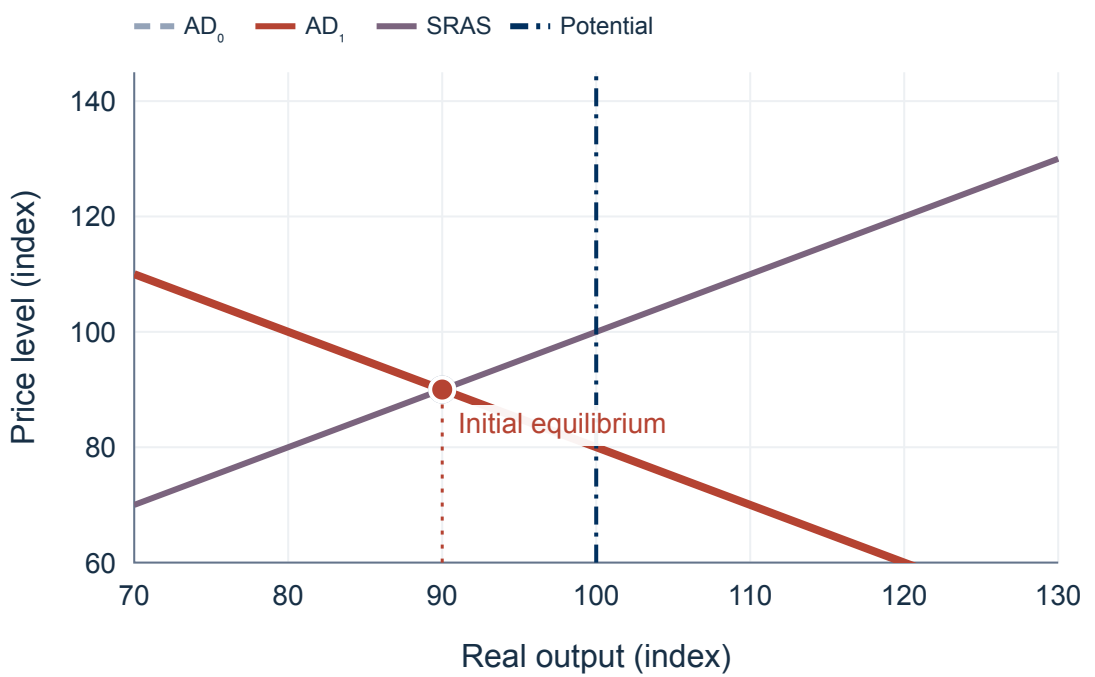
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# Annual Budget Balance and Stabilization

## Demand recession · Starting output = 90 · Potential output = 100

### Contractionary · AD decreases



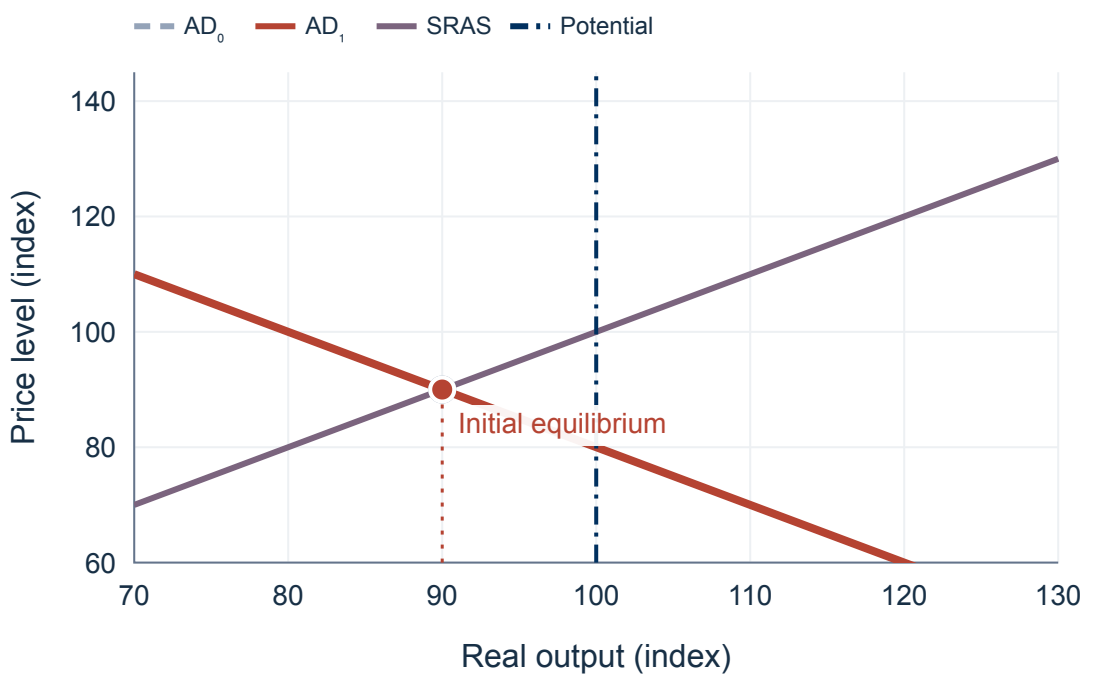

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Output is already below potential.

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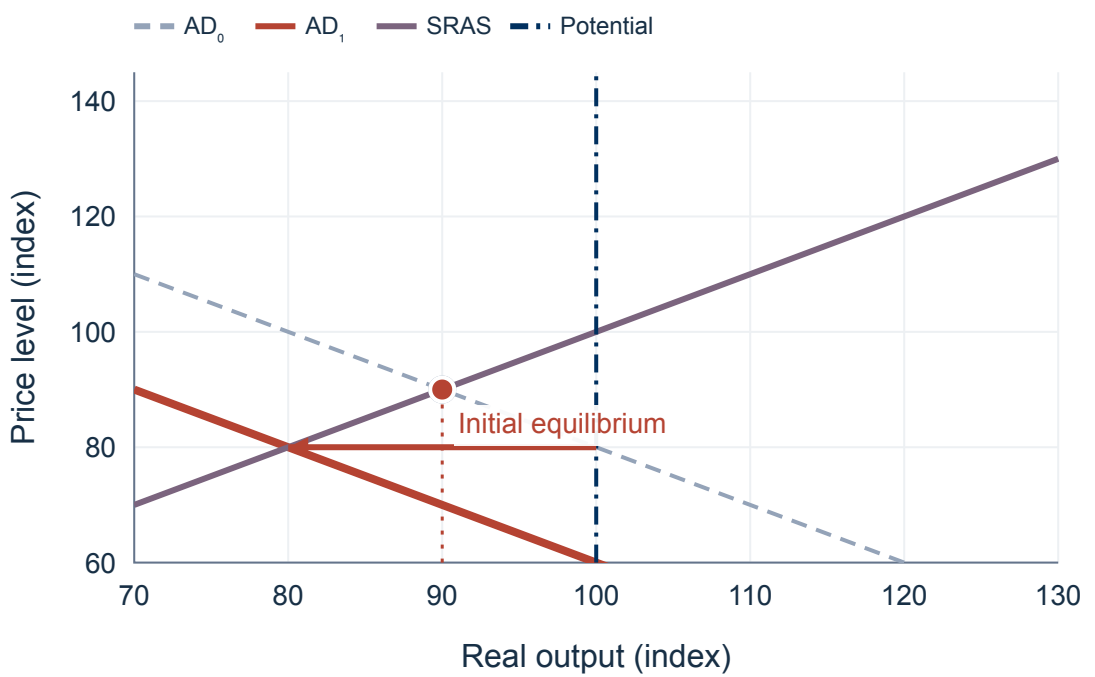
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Falling receipts trigger tax increases or spending cuts.

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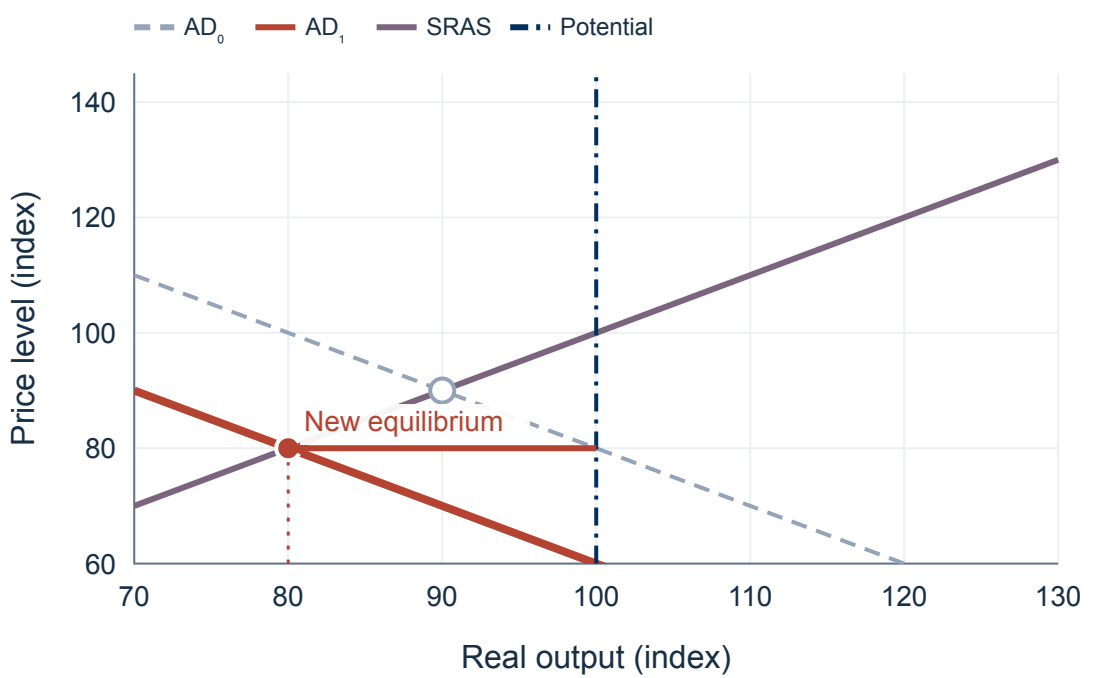
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The offset reduces AD further.

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### Annual-balance requirement

Falling receipts trigger tax increases or spending cuts.

### Demand response

The offset reduces AD further.

### Deeper output gap

Output falls from 90 to 80.

**A strict annual-balance rule can work against automatic stabilizers.**

# Fiscal Stabilization and Sustainability

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Fiscal space allows stabilizers and temporary support to operate.

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### Debt over time

Assess debt relative to GDP, interest costs, growth, and future balances.

A deficit is an annual flow; debt is a stock accumulated over time.

# Fiscal Policy: Final Checkpoint

Explain the mechanism and the evidence you would need.

**A. Income falls; the tax schedule stays unchanged.**

**B. A new law raises weekly unemployment benefits.**

**C. With MPC = 0.8, compare a $100 purchase and a $100 tax cut.**

**D. The deficit rises in a recession. Does this prove new stimulus?**

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Check new laws, automatic stabilizers, and the cyclically adjusted balance.

# Fiscal Policy Cheat Sheet: Demand

Expenditure: C + I + G + NX. Purchases enter G; household taxes and transfers influence C; business taxes can influence I.

| Policy | Main mechanism | Curve | Short-run result |
|---|---|---|---|
| **Increase purchases** | Government buys more current goods and services. | AD right | Output and price level rise. |
| **Cut household taxes** | Disposable income rises; households spend part. | AD right | Output and price level rise. |
| **Increase transfers** | Recipients spend part of the extra income. | AD right | Output and price level rise. |
| **Cut purchases or transfers; raise household taxes** | Spending falls. | AD left | Output and price level fall. |

First three: expansionary. Fourth: contractionary. Transfers themselves are not G.

Accounting: BEA · Demand mechanism: CBO

# Fiscal Policy Cheat Sheet: Supply

Fiscal policy can also affect firms' costs and the economy's productive capacity.

| Policy | Main mechanism | Possible curve shifts | Conditional result |
|---|---|---|---|
| **Productive infrastructure, education or research** | Purchases support demand now; successful investment improves capacity later. | AD right initially; possibly SRAS and LRAS right later. | Potential output can rise. |
| **Lower production taxes or subsidize inputs** | Lower marginal production costs can encourage supply. | Potentially SRAS right. | Output rises and price level falls from the supply effect alone. |

Design and financing matter. Public spending must improve productivity to raise capacity.

Fiscal framework: CBO, How CBO Analyzes Fiscal Policies

# Demand Shocks: Adjustment and Policy

Y* = potential output. Automatic adjustment below assumes no policy response and unchanged potential.

| Initial shock | Immediate result | Automatic adjustment | Fiscal response to the gap | Policy consequence |
|---|---|---|---|---|
| **AD left** | Y < Y*; price level falls. | Wages and costs fall; SRAS moves right. | Expansionary policy moves AD right. | Supports output and offsets some downward price pressure. |
| **AD right** | Y > Y*; price level rises. | Wages and costs rise; SRAS moves left. | Contractionary policy moves AD left. | Returns output toward potential and eases price pressure. |

Self-adjustment changes SRAS through wages and costs; fiscal stabilization usually changes AD.

# Supply Shocks: Adjustment and Policy

Y* = potential output. These two cost shocks leave potential unchanged; adjustment assumes no policy response.

| Initial shock | Immediate result | Automatic adjustment | Fiscal response to the gap | Policy consequence |
|---|---|---|---|---|
| **SRAS left** | Y < Y*; price level rises. | Lower wages or reversal of the cost shock can move SRAS right. | Expansionary policy moves AD right. | Supports output, but raises the price level further. |
| **SRAS right** | Y > Y*; price level falls. | Higher wages or reversal of the favorable shock can move SRAS left. | Contractionary policy moves AD left. | Returns output toward potential, but lowers the price level further. |

After an adverse supply shock, demand policy faces an output–price tradeoff.

# Potential Output and Policy Tradeoffs

### Productive capacity changes

LRAS shifts; SRAS may also shift. Adjustment is toward the new potential output.

### Choose the relevant benchmark

Fiscal choices depend on the remaining gap and the policy objective. Restoring the old GDP level may be inappropriate.

### Keep the two mechanisms distinct

Wage and cost adjustment shifts SRAS; fiscal stabilization usually shifts AD.

Results depend on timing, financing, household spending, productivity, and central-bank responses.